

Data documentation & metadata

TASK -4

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Provenance Notes :

1. Data Provenance & Scope:

This document provides the technical schema and compliance framework for the subset of the **Adult Psychiatric Morbidity Survey (APMS) 2014** used in this research. The data was originally commissioned by NHS Digital and collected by NatCen Social Research. The scope of this documentation covers the aggregated prevalence rates of psychiatric disorders across gender and regional demographics in England.

2. Data History and Summary :

- **Source:** Adult Psychiatric Morbidity Survey (APMS) 2014.
- **The Sample:** It covers adults aged 16 and over living in private households in England.
- **Geographic Scope:** National (England), further broken down by 9 Government Office Regions (GOR).
- **Sample Size:** Approximately **7,500** individual interviews (represented as aggregated prevalence percentages in this specific CSV).
- **Granularity:** Aggregated (High-level summary statistics, not individual patient records).
- **Dataset size :** The dataset includes **702 records** and **15 variables**.

3. Collection Method :

- **Methodology:** Face-to-face interviews using **CAPI** (Computer-Assisted Personal Interviewing) and self-completion sections for sensitive topics (like drug use or self-harm).

- **Sampling:** A **stratified probability sample**. They used a mathematical system to make sure the 7,500 people interviewed represented all of England accurately.

3. Data Cleaning & Transformation Log :

- **Filtering:** "Removed rows containing 'All Adults' to focus on gender comparisons."
- **Handling Nulls:** "Rows with missing prevalence scores were excluded from the skewness calculation."
- **Recoding:** "Grouped AUDIT scores into 'Low Risk' and 'High Risk' categories for better visualization."

4. Metadata (The Data Dictionary) :

Variable Name	Type	Description
Disorder_Categories	Categorical	High-level groupings (e.g., Alcohol Use, PTSD, ADHD).
Disorder	Categorical	The specific condition or scale threshold (e.g., AUDIT 20+).
Sex	Categorical	Population segment (Men, Women, All Adults).
Geographic_area	Categorical	The UK region where the data was collected.
Prevalence_Observed_%	Numeric	The raw percentage of the sample meeting clinical criteria.

Variable Name	Type	Description
Prevalence_AgeStd_%	Numeric	Percentage adjusted to the standard UK age structure.
Sample_size	Numeric	Total number of participants in that specific segment.
CI_Lower_95_%	Float	The lower bound of the 95% Confidence Interval. We are 95% sure the true population value is not below this.
CI_Upper_95_%	Float	The upper bound of the 95% Confidence Interval. We are 95% sure the true population value is not above this.
CI_Width_pp	Float	The width of the Confidence Interval in percentage points. A smaller width means higher precision.
Standard_error%	Float	Measures the "standard deviation" of the sample mean. Lower values indicate a more stable estimate.
Region_Population	Numeric	Total region population
Population_Weight	Float	Total population weight
Source_Table	Float	Gives details about which table was used

Compliance Note :

5. NHS data usage & Compliance Checklist :

To ensure adherence to the **UK Data Protection Act (2018)** and **GDPR**, the following compliance measures have been verified:

Compliance Requirement	Verification Status	Evidence / Notes
Anonymization Standard	Verified	Data is provided as aggregated percentages. No individual-level identifiers (names, DOB, NHS numbers) are present.
GDPR Lawful Basis	Verified	Processing is conducted under Article 6(1)(e) : "Public Task" and Article 9(2)(j) : "Scientific or Historical Research Purposes."
Confidentiality (Small Cell Counts)	Verified	In accordance with NHS Digital protocol, any demographic cross-section with fewer than 5 respondents was suppressed in the source to prevent indirect re-identification.
Data Minimization	Verified	Only the variables relevant to psychiatric prevalence and demographic distribution were processed for this research.
Purpose Limitation	Verified	The dataset is used strictly for academic statistical analysis; no commercial profiling or marketing activities were conducted.

Compliance Requirement	Verification Status	Evidence / Notes
Integrity & Security	Verified	The dataset is stored on a local, password-protected machine and is not hosted on public, unencrypted cloud servers.
Source Citation	Verified	All data used is credited to the original owners: NHS Digital and NatCen Social Research.

This dataset is classified as **Secondary Aggregated Data**, which falls outside the scope of 'Personal Data' under GDPR because it is impossible to identify a specific human being from these percentages.

For example, instead of seeing "John Doe has ADHD," we only see "10% of Men in London have ADHD."